

Software Requirement Specification (SRS)

Project: Ultimate Customizable Product Platform

1.Introduction

This platform lets you easily design custom items like T-shirts, mugs, and stickers while watching the price update in real time. Its standout feature is a secure "Open Delivery" protocol that lets you unbox and check the print quality right in front of the delivery agent. If you are happy with the product, you share a secure OTP to accept it; if it is damaged or incorrect, you can reject it instantly at your doorstep. This creates a completely risk-free shopping experience that builds unmatched customer trust.

1.1 Purpose

This document specifies the software requirements for the Ultimate Customizable Product Platform. The platform allows users to design, customize, and purchase a wide variety of merchandise (e.g., T-shirts, stickers, home decor) while featuring a transparent "Open Delivery" system to ensure customer trust.

1.2 Scope

The system consists of:

- An interactive Design Studio (Web/Mobile).
 - An E- commerce Marketplace for catalog browsing and purchasing.
 - An Open Delivery Module allowing customers to verify products upon delivery.
 - A Merchant/Admin Dashboard for order fulfillment and product management.
- 2.Overall Description

2.1 User Classes and Characteristics

- End Customer: Browses products, uses the customization tool, places orders, and tracks deliveries.
- Vendor/Printer (Optional/Scalable): Receives the raw high- resolution

design files to print and ship the products.

- **Delivery Partner:** Handles shipping and facilitates the "open delivery" inspection protocol.
- **System Administrator:** Manages users, base product catalogs, payment gateways, and disputes.

2.2 Design and Implementation Constraints

- **Scalability:** High storage capability for user- uploaded high- definition images/vectors.
- **Performance:** The customization canvas must render smoothly without lagging on mobile devices.
- **Security:** Compliance with PCI- DSS for payments and secure handling of user data.

3. System Features & Functional Requirements

3.1 Interactive Product Design Studio

- **Base Product Selection:** Users can choose a blank canvas product (Tshirt, sticker sheet, mug, etc.), select variants (size, color, material), and see a real-time 2D/3D preview.
- **Custom Asset Upload:** Users can upload images (.png, .jpg, .svg).
The system must assess image resolution and warn users if the quality is too low for printing.
- **Text & Graphic Manipulation:** Users can add text, choose fonts, change colors, resize, rotate, layer elements, and use pre- loaded clipart/templates.
- **Dynamic Pricing:** The price must update dynamically based on customization factors (e.g., printing on front vs. front + back, size of the design).

3.2 E- Commerce & Checkout System

- **Shopping Cart:** Save custom configurations, update quantities, and display breakdowns of base price, customization fees, taxes, and shipping.

- **Secure Payment Gateway:** Integration with multiple payment options (Credit/Debit cards, UPI, Wallets, and Cash on Delivery).
- **Order Management:** Generation of a unique Order ID and automated creation of a print- ready file package (e.g., high- res transparent PDF/PNG + mock-up placement coordinates).

3.3 The "Open Delivery" System

- **Definition:** A trust- building protocol where the delivery agent allows the customer to open the package and inspect the custom print quality before completing the transaction (highly critical for Cash on Delivery).
- **Verification Protocol:** * The delivery agent app generates a Verification OTP or barcode.
 - If the product matches the design and is undamaged, the customer provides the OTP to accept.
- **Instant Rejection/Return Module:** If the print is defective, incorrect, or damaged, the delivery agent marks it as "Rejected at Doorstep" via their app, triggering an automated refund or reprint process instantly.

4.Non- Functional Requirements

4.1 Performance Requirements

- The Design Studio canvas must respond to user inputs (scaling, dragging) within 100ms.
- Print file generation should happen asynchronously in the background so it doesn't slow down the user's checkout experience.

4.2 Security Requirements

- All user uploads must be scanned for malicious scripts.
- Customer design IP (Intellectual Property) must be protected; designs should not be accessible by other users unless explicitly shared.

4.3 Reliability & Availability

- The system should maintain 99.9% uptime, especially during peak

holiday shopping seasons.

5.System Architecture & Technical Stack

- Recommended Layer Technology
- Front- End
- React.js / Next.js or Flutter
- Design
- Fabric.js / Three.js Canvas
- Back- End
- Node.js (Express) or Python (Django)
- Purpose
- Provides a smooth, responsive UI across web and mobile.
- Powerhouse libraries for 2D/3D HTML5 canvas manipulation.
- Handles heavy API loads, order processing, and logic efficiently.
- Database PostgreSQL +
- MongoDB
- Storage AWS S3 or Google
- Cloud Storage

6.Data Flow: From Design to Open Delivery

[User Designs Product]< [Dynamic Price Calculated]< [Checkout & Payment]

[Open Delivery Acceptance]< [Agent Delivers & Inspects] <[Print File Generated & Ship